

Hiltown Hotel Energy Management and Automation System

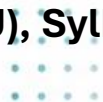
Taltola, Sylhet-3100, Bangladesh.

Proposed By:



Department of Electrical and Electronic Engineering (EEE),
RTM Al-Kabir Technical University (RTM-AKTU), Sylhet.

7th September, 2026.



About Us

The **HilTown Hotel Energy Management and Automation System** project is presented by the **EEE Department of RTM Al-Kabir Technical University, Sylhet** consisting of **experienced (Former IoT Support Engineer, ANTT Robotics, Dhaka) research faculty members and students** with expertise in energy systems, automation, IoT, and intelligent control technologies.

The faculty members are actively engaged in different **advanced research activities** and have been recognized in numerous research works under different **international journals and conferences**, in this domain some of them including:

- *A Smart Medicine Reminder Kit with Mobile Phone Calls and Some Health Monitoring Features for Senior Citizens*, doi: [10.1016/j.heliyon.2024.e26308](https://doi.org/10.1016/j.heliyon.2024.e26308)
- *A Smart Multi-User Wireless Nurse Calling System and E-notice Board for Health Care Management*, doi: [10.1007/978-981-16-7597-3_35](https://doi.org/10.1007/978-981-16-7597-3_35)
- *Development of an Intravenous Fluid Monitoring, Warning, and Reverse Flow Blocking System*, doi: [10.1109/WIECON-ECE57977.2022.10150612](https://doi.org/10.1109/WIECON-ECE57977.2022.10150612)
- *AgriCall: An IoT-Integrated GSM-Based Irrigation Control System for Sustainable Agriculture*.
- *An IoT-Based Smart Adaptive Highway Management System for Sound Pollution Reduction and Traffic Control*, doi: [10.1109/QPAIN66474.2025.11172235](https://doi.org/10.1109/QPAIN66474.2025.11172235)
- *IoT-Based Smart Waste Management System Using NodeMCU and Cloud Computing, IoT-Based Smart Waste Management System Using NodeMCU and Cloud Computing*, doi: [10.1109/QPAIN66474.2025.11171905](https://doi.org/10.1109/QPAIN66474.2025.11171905)
- *IoT and Mobile App-Based Real-Time Remote E-Health Care System for Rural Communities in Bangladesh*, doi: [10.1109/QPAIN66474.2025.11171985](https://doi.org/10.1109/QPAIN66474.2025.11171985)

Now they find scopes in developing innovative engineering solutions and combining academic research with real-world industrial applications.



Introduction

Hiltown Hotel– Smart Room Management System

Hiltown Hotel is a modern hotel environment where efficient room management, security, and energy utilization are essential for smooth daily operations.

The proposed Smart Room Management System integrates RFID access control, IoT-based appliance control, human presence detection, door monitoring, and energy monitoring into a centralized platform.

The system is designed to give hotel management remote visibility and control over individual rooms, while reducing unnecessary appliance usage and improving overall energy efficiency.



KEY FOCUS AREAS

Access Control

Room Automation

Energy Management

Remote Monitoring

Problem and Proposed Solution

THE CHALLENGE

Uncontrolled appliance usage — AC, fan, heater and lighting may remain ON when rooms are unattended.

Limited remote visibility — Management cannot easily determine the real-time status of room appliances.

No role-based appliance control — Housekeeping may unintentionally use high-consumption appliances during room servicing.

Manual room monitoring — Door status and room occupancy require physical checking.

Energy consumption is difficult to track — Limited real-time and historical information makes energy optimization difficult.

Conventional switchboards — Standard mechanical switches do not support centralized IoT-based control.

OUR PROPOSED SOLUTION

RFID-based authorization separates Guest and Housekeeping access.

Smart relay control enables both wall-switch and remote appliance control.

Human presence detection identifies prolonged room vacancy.

Automatic Power Saving Mode switches OFF unnecessary loads after a defined period of inactivity.

Door monitoring provides real-time open/closed status.

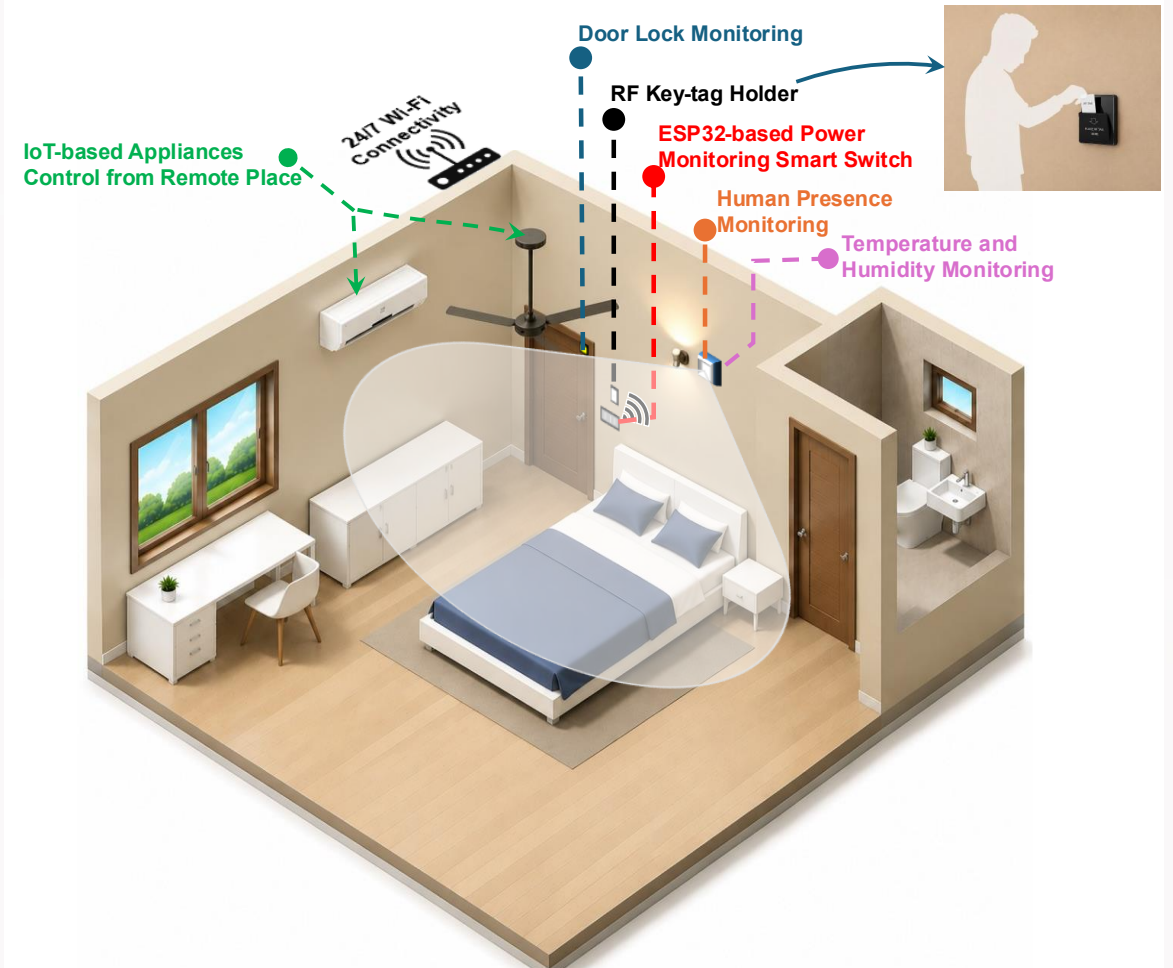
Energy monitoring tracks voltage, current, power and kWh consumption.

Centralized dashboard provides room-wise monitoring, control and historical energy data.

Project Overview

The proposed system is an IoT-based hotel room automation solution developed for “Hiltown-Hotel” to improve room-level control, security, operational efficiency, and energy management. The system integrates RFID access control, smart appliance control, human presence detection, door-status monitoring, and real-time energy monitoring into a centralized management platform.

- 01 Identify** RFID technology identifies whether the room is accessed by a Guest or Housekeeping staff.
- 02 Authorize** Appliance permissions are automatically assigned according to the user’s access category.
- 03 Monitor** ESP32-based controllers collect information from occupancy, door, voltage, and current sensors.
- 04 Control** Hotel management can remotely monitor and control room appliances through the central dashboard.
- 05 Optimize** When a guest is absent for a predefined period, the system activates Power Saving Mode and switches off unnecessary loads.



RFID Access



ESP32
Controller



Relay &
Sensors



IoT/Firebase



Management
Dashboard

Benefits

Energy Efficiency

Reduces unnecessary consumption through occupancy-based automation and remote load control.

Operational Control

Provides management with centralized, room-wise monitoring and appliance control.

Improved Security

RFID authorization and door-status monitoring provide better control over room access and activity.

Reduced Operational Waste

Limits unnecessary use of high-power appliances by housekeeping staff and unattended rooms.

Data-Driven Management

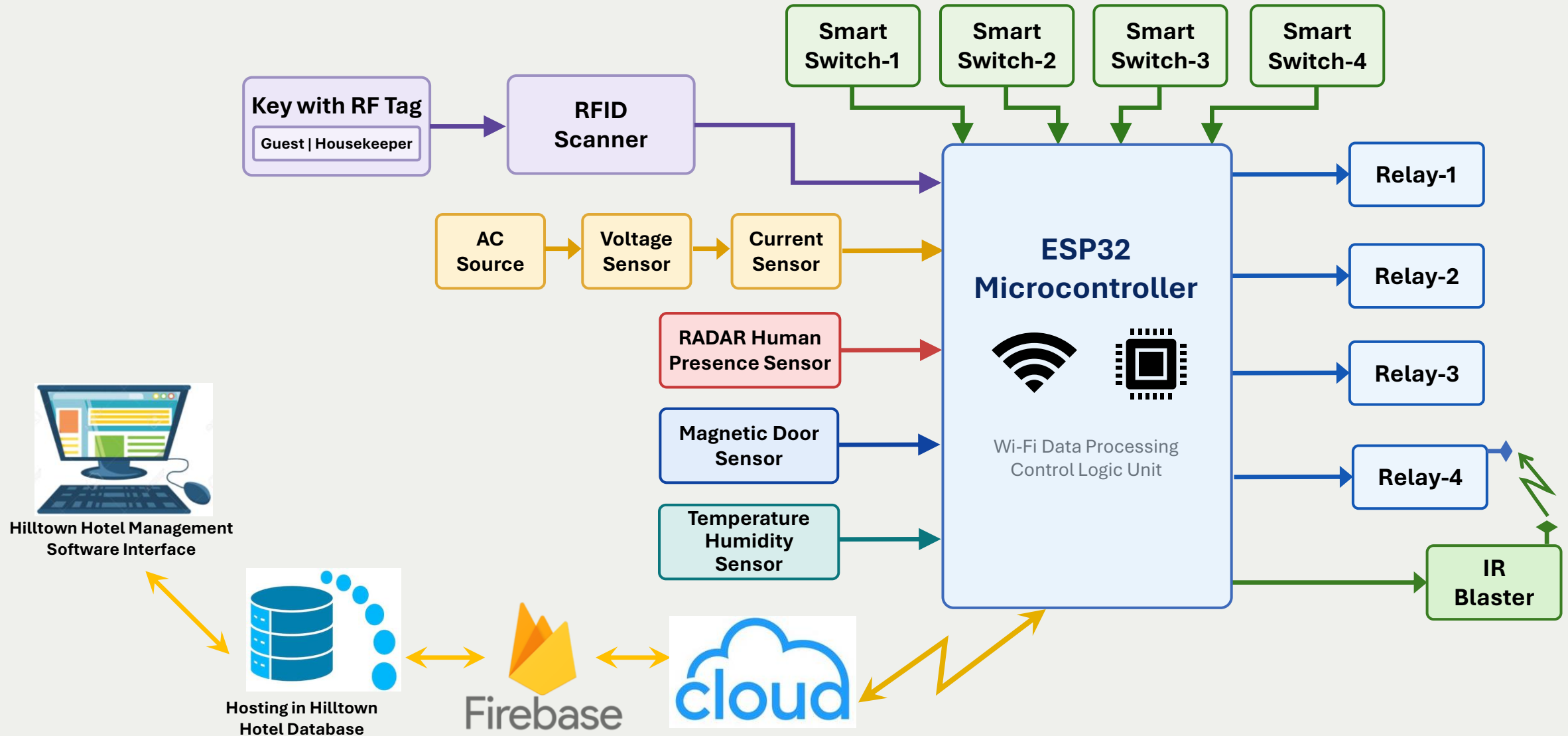
Historical energy data helps management analyze consumption patterns and identify opportunities for optimization.

Scalable Architecture

The system can be extended to additional rooms and integrated with existing hotel management infrastructure.



Approach and Methodology



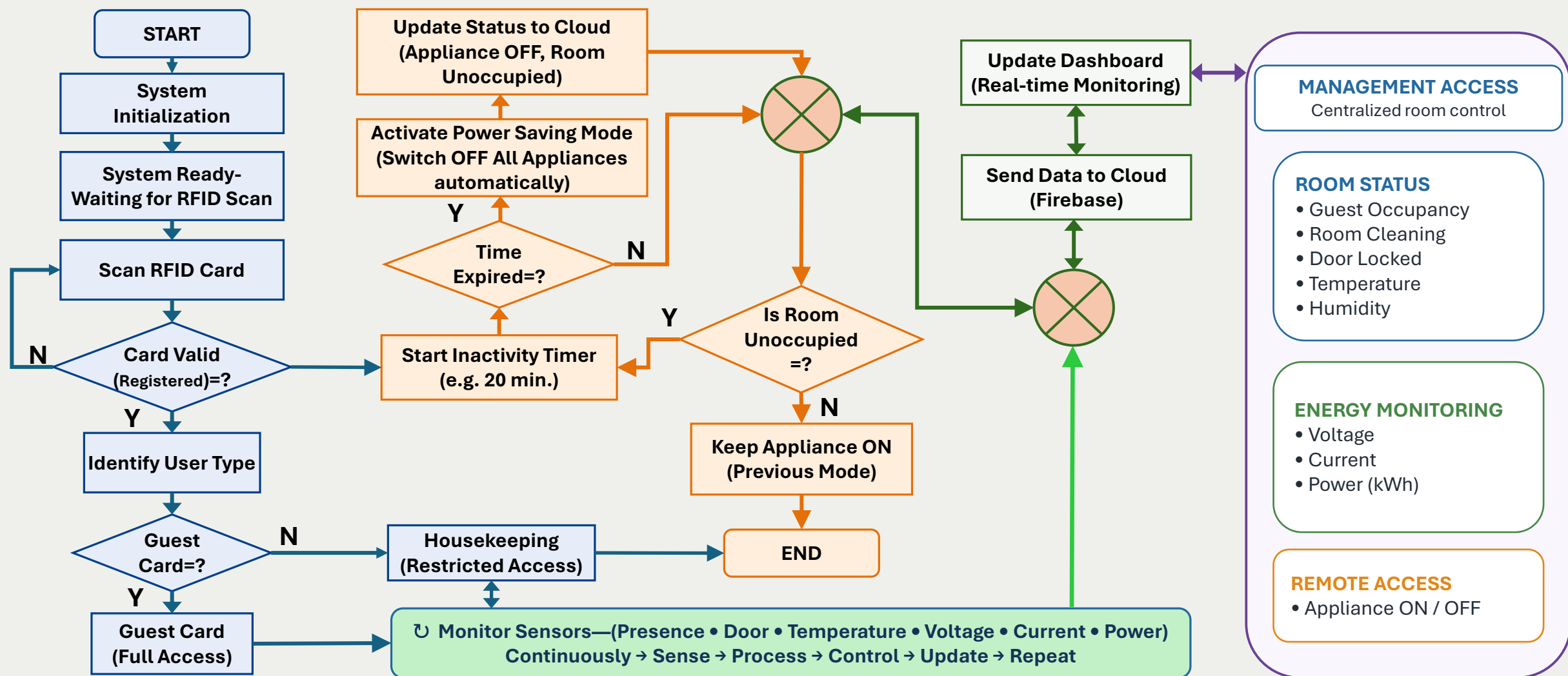
Project Details

1. ACCESS & AUTHORIZATION

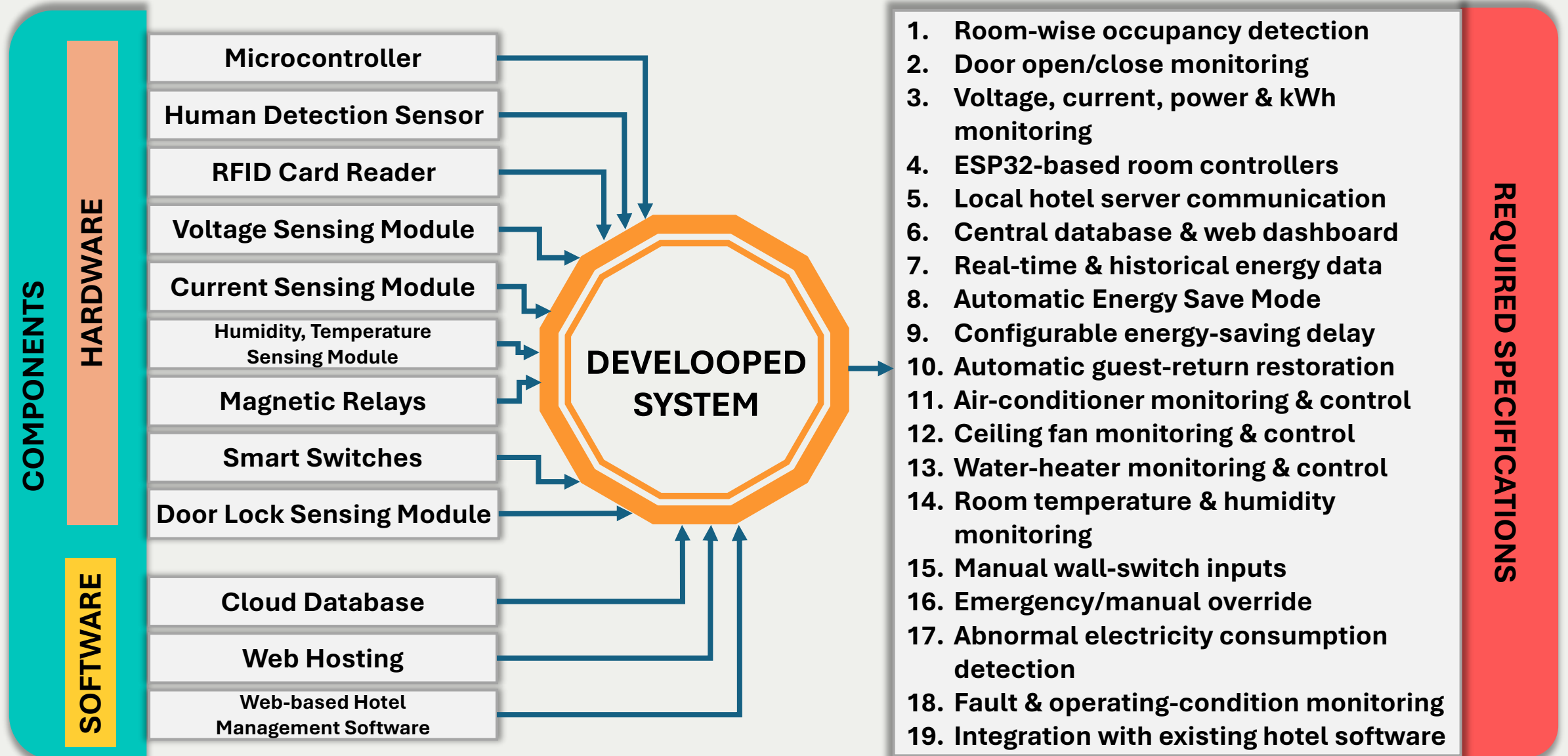
2. ENERGY MANAGEMENT

3. MONITORING & DATA FLOW

4. MANAGEMENT DASHBOARD



System Components



System Prototype

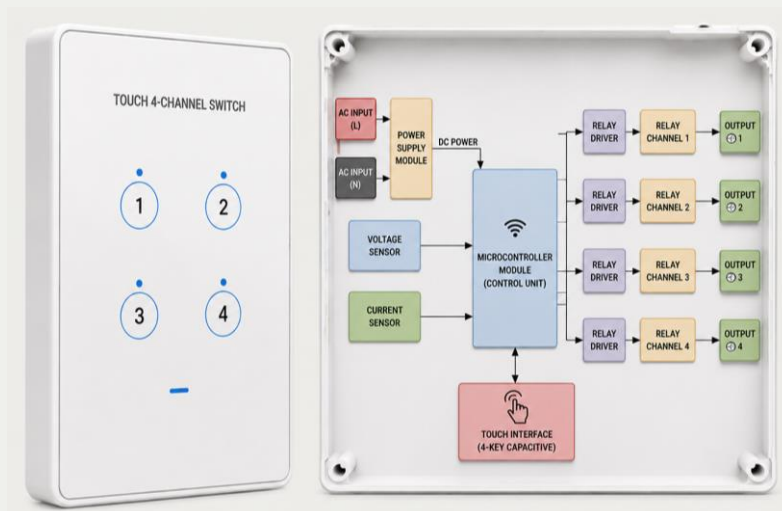


Fig. 1. Smart IoT Switch



Fig. 2. Human Presence, Temperature, Humidity Module



Fig. 3. RF Key Reader

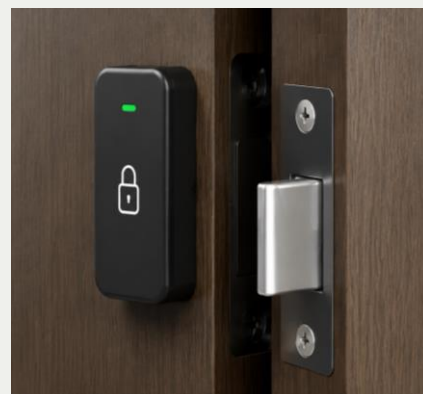


Fig. 4. Door-lock Sensing Module

Hiltown Hotel Energy Management and Automation System Dashboard

Room 5-B-1

Room Temp.: 24°C
Room Humidity: 80%
Room Status: LOCKED
Room Mode: Servicing
Power Consumption: 0.44 KWh
Load Current: 2A

Room Appliances

Light:	ON	ON	OFF
Fan:	OFF	ON	OFF
AC:	ON	ON	OFF
Heater:	OFF	ON	OFF

Room 5-B-2

Room Temp.: 20°C
Room Humidity: 72%
Room Status: UNLOCKED
Room Mode: Guest
Power Consumption: 0.33 KWh
Load Current: 1.5A

Room Appliances

Light:	OFF	ON	OFF
Fan:	ON	ON	OFF
AC:	ON	ON	OFF
Heater:	OFF	ON	OFF

Specifications:

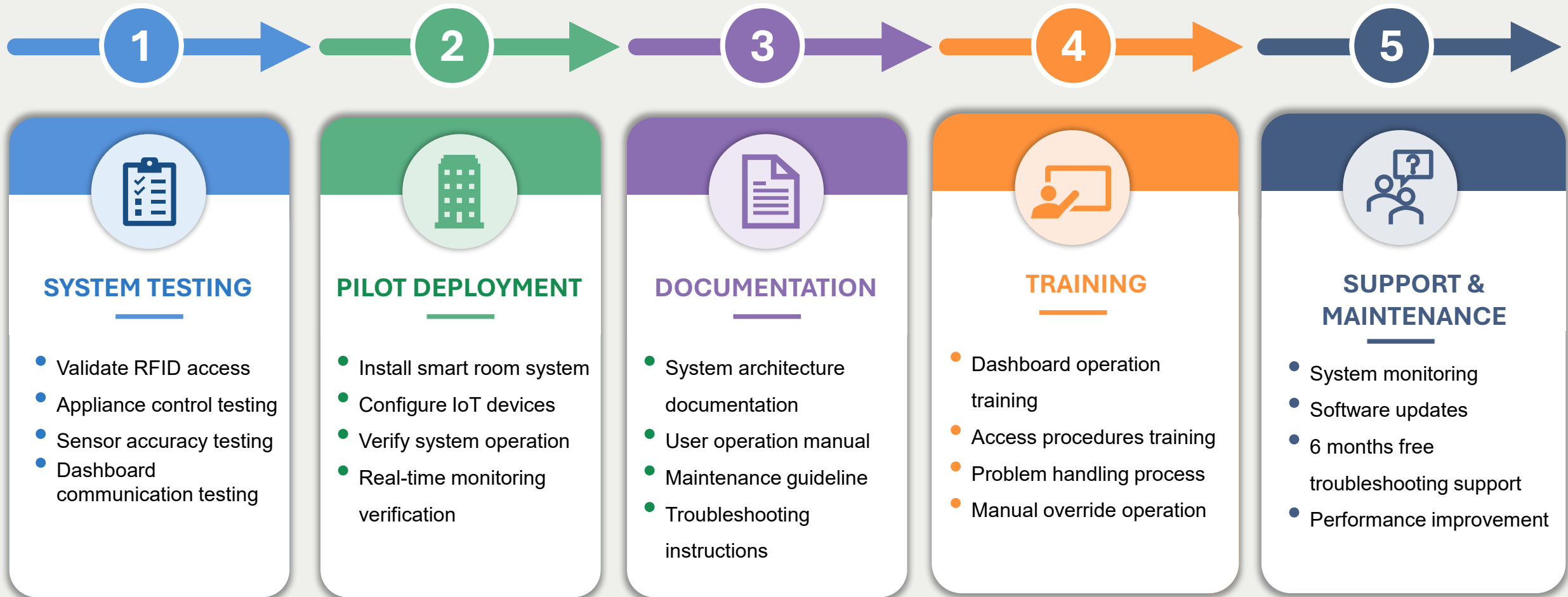
Input voltage: AC 220V, 50Hz
Input Current: 2A
Required Connectivity: Wi-Fi
Coverage: 10 meter, 120°
Temperature: 0°C – 50°C

**Device design may vary after developing.*

Timeline (Gantt chart)

[illegible]

Implementation Roadmap



Market Overview

Required Capability	ZKTeco	Sonoff	Tuya Presence	SmartX	Tuya Energy Meter	Our System
Manual Room Control	✓	✓	✗	✗	✗	✓
Web / Remote Control	✗	✓	✓	✓	✓	✓
RFID Access	✓	✗	✗	✗	✗	✓
Guest / Housekeeping Roles	✗	✗	✗	✗	✗	✓
Human Presence Detection	✗	✗	✓	✗	✗	✓
Door Monitoring	✗	✗	✗	✗	✗	✓
Temperature / Humidity	✗	✗	✗	✓	✗	✓
Energy Monitoring	✗	✗	✗	✗	✓	✓
Automatic Energy Saving	Limited	✗	Trigger	✗	✗	✓
Centralized Dashboard	✗	✗	✗	✗	✗	✓
Historical Energy Data	✗	✗	✗	✗	Limited	✓
Hotel-specific Customization	✗	✗	✗	✗	✗	✓
Price (BDT)	2800	2900	3350	3000	3500	8500-9000

SWOT Analysis



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STRENGTHS

- All-in-one smart room automation platform
- RFID-based access control
- Real-time energy monitoring
- Remote appliance control
- Automatic energy saving



W

WEAKNESSES

- Initial installation complexity
- Requires IoT infrastructure
- Wi-Fi/cloud dependency



O

OPPORTUNITIES

- Guest satisfaction
- Growing smart hotel market
- Expansion to multiple rooms/hotels
- Energy cost reduction demand
- Sustainable building solutions



T

THREATS

- Existing automation competitors
- Adoption cost barrier
- Cybersecurity concerns
- Maintenance challenges

Budget

Description	No. of Units	Rate	Total Cost (BDT)
Direct Cost			
Materials and Components	5	8500	42,500
Wiring, Connectors and Customised Casing	5	1400	7,000
Research and Development (Hardware & Software)	1	8000	8,000
Honorarium	1	20000	20,000
A. Sub Total Direct Cost			77,500
Other Cost			
Local transport while First Testing and Installation on Spot	2	400	800
Local transport while Final Testing and Installation on Spot	2	400	800
Installation cost (Electrician Cost)	5	1000	5,000
Miscellaneous	Lump sum		2,857
B. Sub Total Other Cost			9,457
Institutional Overheads for Full Facility Lab Use (10% of (A+B))			8,696
C. Sub Total Institutional Overheads Cost			8,696
Total Cost (A+B+C)			95,652
VAT (15%)			14,348
Total Estimated Cost			110,000

Project Summary

THE PROBLEM



High energy wastage in unoccupied rooms



Manual operation and inefficient operations



Unmonitored appliance usage



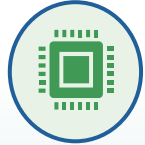
Security and access management issues

SYSTEM ARCHITECTURE



INPUT DEVICES

- RFID Reader
- Occupancy Sensor
- Door Sensor
- Temperature Sensor
- Energy Meter



CONTROLLER

ESP32 Microcontroller

- Data Collection
- Local Processing
- Device Control



CONNECTIVITY

Wi-Fi Connection

- Secure Data Transfer
- Real-time Communication



CLOUD PLATFORM

Firebase Cloud

- Data Storage
- User Management
- Real-time Database



APPLICATION

Web Dashboard Management

- Real-time Monitoring
- Control from Remote
- Reports and Analysis



THE SOLUTION



IoT based smart room automation



Real-time monitoring and remote control



Energy optimization and cost savings



Enhanced security and guest satisfaction

PROJECT IMPACT

30-40%

Reduce unnecessary power consumption and operational cost.

Better

Guest experience in smart environment for a more comfortable stay.

Smart Security

Controlled access and real-time door monitoring.

Centralized

Management and automated control from remote place saves time and effort.

Sustainable Future

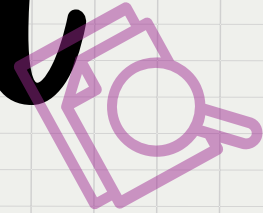
Environment friendly solution for modern smart hotel.



Thank You



Thank you for supporting our
innovative ideas.



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